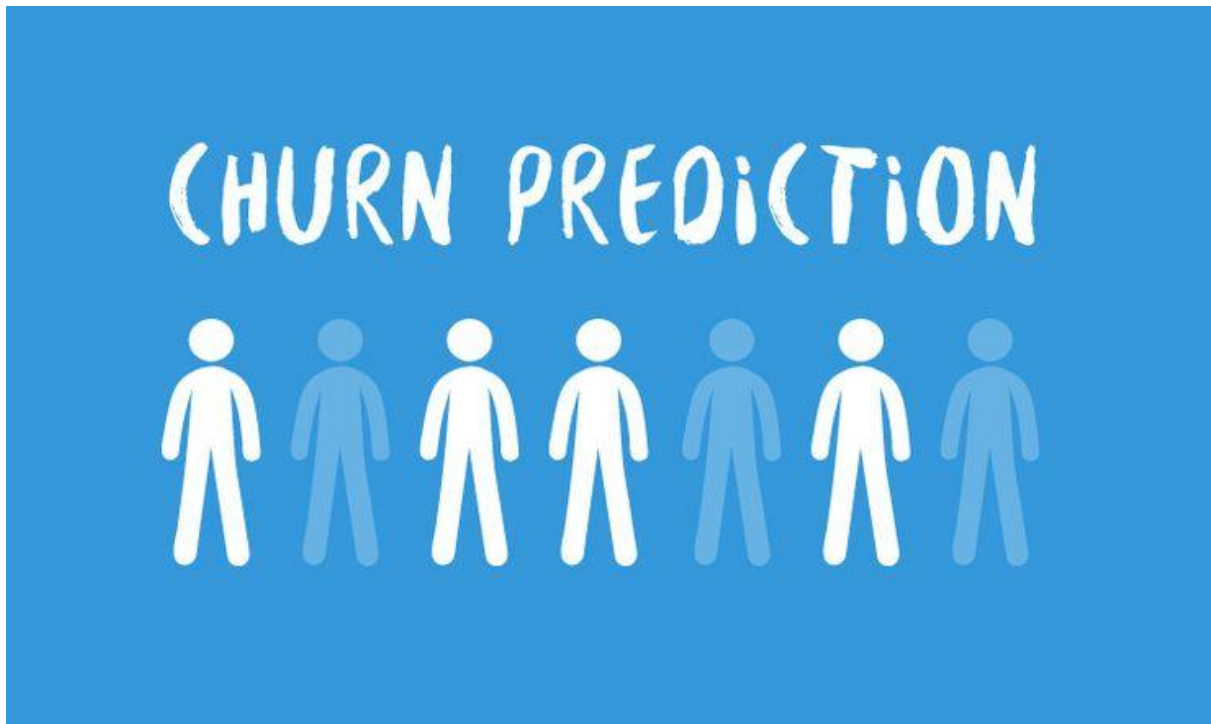


Project Title:

Customer Churn Analysis and Predictive Modelling

A data-driven analysis project focused on understanding customer behaviour to identifying churn patterns and support retention strategies through Logistic Regression.



Tools & Technologies Used:

Python, Pandas, Matplotlib, Seaborn, Scikit-learn

~ Prepared By:

Sayantani Dalui

Problem Statement

Customer churn is a major challenge for subscription-based and service-oriented businesses. When customers discontinue their services, it directly impacts revenue, profitability, and long-term growth. Acquiring new customers is significantly more expensive than retaining existing ones, making churn reduction a critical business priority. Therefore, understanding the key factors that influence customer churn and identifying at-risk customers in advance is essential for developing effective retention strategies.

This project aims to analyse customer data to uncover patterns, trends, and relationships associated with churn behaviour. By exploring demographic attributes, service usage details, contract types, billing information, and support services, the objective is to determine which factors contribute most significantly to customer attrition. In addition to exploratory data analysis, predictive modelling is applied to estimate the likelihood of churn, enabling proactive business decision-making.

The analysis focuses on answering the following key questions:

- What is the overall churn rate of customers?
- How does customer tenure influence churn behaviour?
- Does contract type (monthly, yearly, etc.) impact churn probability?
- How do monthly charges and total charges relate to churn?
- Is there any relationship between gender and churn?
- Do customers with dependents churn less frequently?
- Are senior citizens more likely to churn?
- How do payment methods affect churn behaviour?
- Does having technical support reduce churn rate?

To further support data-driven decision making, a **Logistic Regression model** was developed to estimate the probability of customer churn based on key influencing factors. Since churn is a binary outcome (Yes/No), Logistic Regression was used to analyse the relationship between customer attributes and the likelihood of churn.

Special attention was given to class imbalance in the dataset to ensure that churn customers were correctly identified. The model performance was evaluated using accuracy, recall, F1-score, and confusion matrix, with a strong focus on recall to ensure maximum identification of at-risk customers.

Data Handling and Preprocessing

1. The dataset was loaded using Pandas for analysis and initial inspection of data types and missing values was performed.
2. During data exploration, it was observed that the **TotalCharges** column was incorrectly read as an object data type instead of numeric.
3. Investigation revealed that blank spaces were present in the TotalCharges column, causing it to be interpreted as a string.
4. The dataset was reloaded using `na_values=""` so that blank spaces were treated as missing values. After this correction, the TotalCharges column was automatically converted to float data type.
5. Missing values in the TotalCharges column were handled by replacing them using the formula:
TotalCharges = Tenure × MonthlyCharges, ensuring logical consistency in the dataset.
6. The **SeniorCitizen** column was originally stored as int64 (0 and 1). Since it represents a categorical variable, its data type was converted from integer to object for proper categorical analysis.
7. After these preprocessing steps, the dataset was clean, consistent, and ready for exploratory data analysis and predictive modelling.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneServ	MultipleLi	InternetSe	OnlineSec	OnlineBac	DevicePro	TechSupp	Streaming	Streaming	Contract	PaperlessB	PaymentMethod	MonthlyCha	TotalCharge	Churn
2	7590-VHVEG	Female	0	Yes	No	1	No	No phone	DSL	No	Yes	No	No	No	No	Month-to	Yes	Electronic check	29.85	29.85	No
3	5575-GNVEE	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No	No	No	One year	No	Mailed check	56.95	1889.5	No
4	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to	Yes	Mailed check	53.85	108.15	Yes
5	7795-CFDCW	Male	0	No	No	45	No	No phone	DSL	Yes	No	Yes	Yes	No	No	One year	No	Bank transfer (auto)	42.3	1840.75	No
6	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	No	No	No	Month-to	Yes	Electronic check	70.7	151.65	Yes
7	9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	Month-to	Yes	Electronic check	99.65	820.5	Yes
8	1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No	Yes	No	Yes	Month-to	No	Credit card (automa)	89.1	1949.4	No
9	6713-OKOMC	Female	0	No	No	10	No	No phone	DSL	Yes	No	No	No	No	No	Month-to	No	Mailed check	29.75	301.9	No
10	7892-POOKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	No	Yes	Yes	Yes	Yes	Month-to	Yes	Electronic check	104.8	3046.05	Yes
11	6388-TABGU	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No	No	No	No	One year	No	Bank transfer (auto)	56.15	3487.95	No
12	9763-GRSKD	Male	0	Yes	Yes	13	Yes	No	DSL	Yes	No	No	No	No	No	Month-to	Yes	Mailed check	49.95	587.45	No
13	7469-LKBCI	Male	0	No	No	16	Yes	No	No	No	No	No	No	No	No	Two year	No	Credit card (automa)	18.95	326.8	No
14	8091-TTVAX	Male	0	Yes	No	58	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	One year	No	Credit card (automa)	100.35	5681.1	No
15	0280-XUGEX	Male	0	No	No	49	Yes	Yes	Fiber optic	No	Yes	Yes	No	Yes	Yes	Month-to	Yes	Bank transfer (auto)	103.7	5036.3	Yes
16	5129-JLPIS	Male	0	No	No	25	Yes	No	Fiber optic	Yes	No	Yes	Yes	Yes	Yes	Month-to	Yes	Electronic check	105.5	2686.05	No
17	3655-SNIQZ	Female	0	Yes	Yes	69	Yes	Yes	Fiber optic	Yes	Yes	Yes	Yes	Yes	Yes	Two year	No	Credit card (automa)	113.25	7895.15	No
18	8191-KWZSG	Female	0	No	No	52	Yes	No	No	No	No	No	No	No	No	One year	No	Mailed check	20.65	1022.95	No
19	9959-WOFKT	Male	0	No	Yes	71	Yes	Yes	Fiber optic	Yes	No	Yes	No	Yes	Yes	Two year	No	Bank transfer (auto)	106.7	7382.25	No
20	4190-MFLUW	Female	0	Yes	Yes	10	Yes	No	DSL	No	No	Yes	No	No	No	Month-to	No	Credit card (automa)	55.2	528.35	Yes
21	4183-MYRFB	Female	0	No	No	21	Yes	No	Fiber optic	No	Yes	Yes	No	No	Yes	Month-to	Yes	Electronic check	90.05	1862.9	No
22	8779-QRDMV	Male	1	No	No	1	No	No phone	DSL	No	No	Yes	No	No	Yes	Month-to	Yes	Electronic check	39.65	39.65	Yes
23	1680-VDCWW	Male	0	Yes	No	12	Yes	No	No	No	No	No	No	No	No	One year	No	Bank transfer (auto)	19.8	202.25	No
24	1066-JKSGK	Male	0	No	No	1	Yes	No	No	No	No	No	No	No	No	Month-to	No	Mailed check	20.15	20.15	Yes
25	3638-WEABW	Female	0	Yes	No	58	Yes	Yes	DSL	No	Yes	No	Yes	No	No	Two year	Yes	Credit card (automa)	59.9	3505.1	No
26	6322-HRFFA	Male	0	Yes	Yes	49	Yes	No	DSL	Yes	Yes	No	Yes	No	No	Month-to	No	Credit card (automa)	59.6	2970.3	No
27	6865-JZNYO	Female	0	No	No	30	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to	Yes	Bank transfer (auto)	55.3	1530.6	No

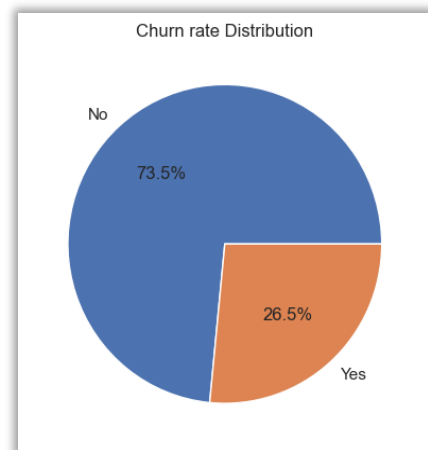
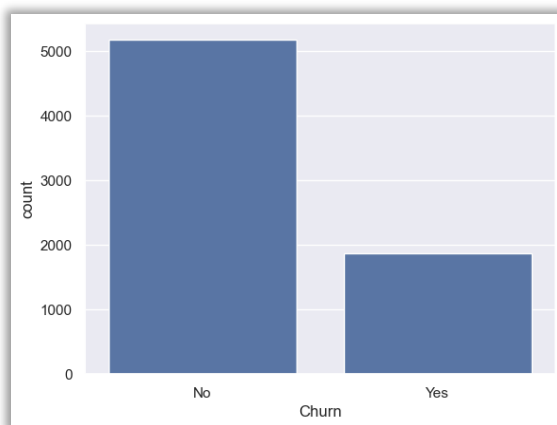
Exploratory Data Analysis (EDA)

1. Overall Churn Distribution

- The overall churn rate is **26.5%**, while **73.5%** of customers did not churn.
- This indicates a class imbalance in the dataset, with non-churn customers forming the majority.

→ i) `sns.countplot(x='Churn', data=customer1)`

ii) `churn_rate=customer1['Churn'].value_counts()
plt.pie(churn_rate,labels=churn_rate.index,autopct='%1.1f%%')
plt.title("Churn rate Distribution")`

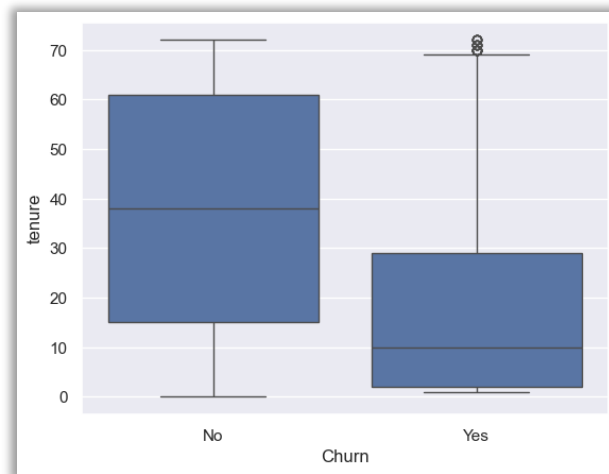


2. Tenure vs Churn

- Customers who churned have significantly lower tenure compared to retained customers.
- The median tenure for churned customers is much lower, indicating that customers are more likely to leave in the early stages of their subscription.
- Long-term customers show higher retention stability.

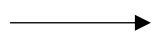


```
sns.boxplot(x='Churn',y='tenure', data=customer1)
```

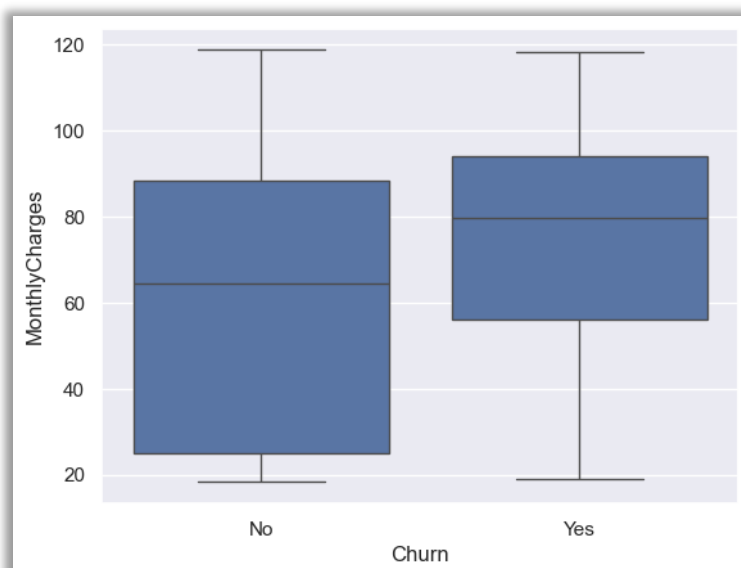


3. Monthly Charges vs Churn

- Customers who churned tend to have higher monthly charges.
- The median monthly charge for churned customers is higher than for non-churn customers.
- Higher pricing plans may contribute to increased churn probability.



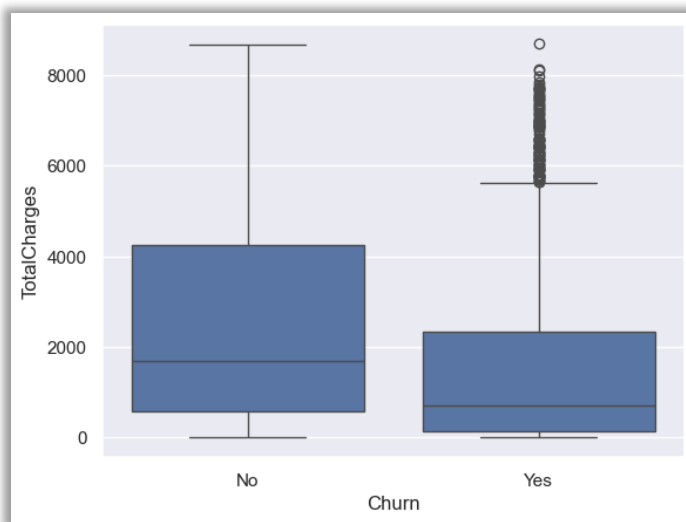
```
sns.boxplot(x='Churn',y='MonthlyCharges', data=customer1)
```



4. Total Charges vs Churn

- Non-churn customers show significantly higher total charges due to longer tenure.
- Churned customers have lower total charges, reflecting shorter customer lifetime.

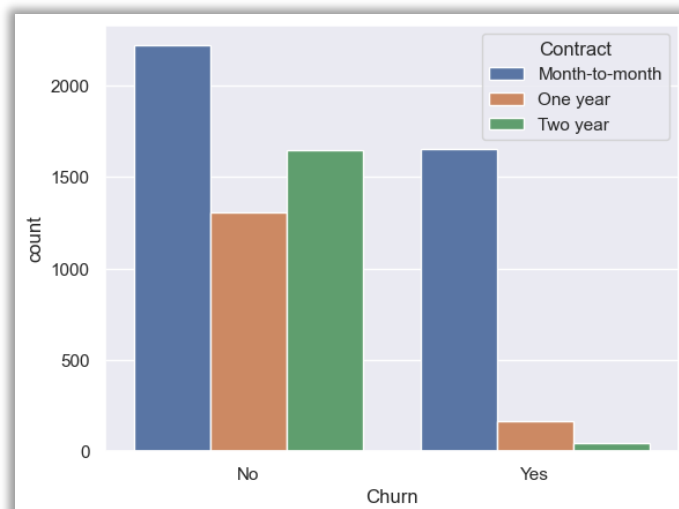
→ `sns.boxplot(x='Churn', y='TotalCharges', data=customer1)`



5. Contract Type vs Churn

- Month-to-month contract customers show the highest churn rate.
- One-year and two-year contracts have significantly lower churn rates.
- Longer contract commitments are associated with stronger customer retention.

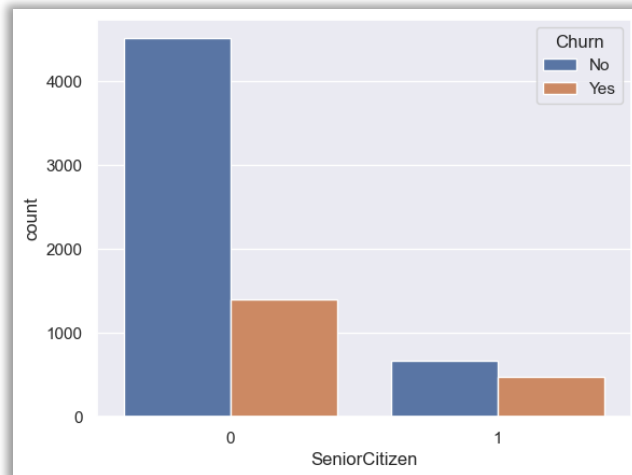
→ `sns.countplot(x='Churn', data=customer1, hue="Contract")`



6. Senior Citizen vs Churn

- Senior citizens have a higher churn proportion compared to non-senior customers.
- This suggests age-related factors may influence retention.

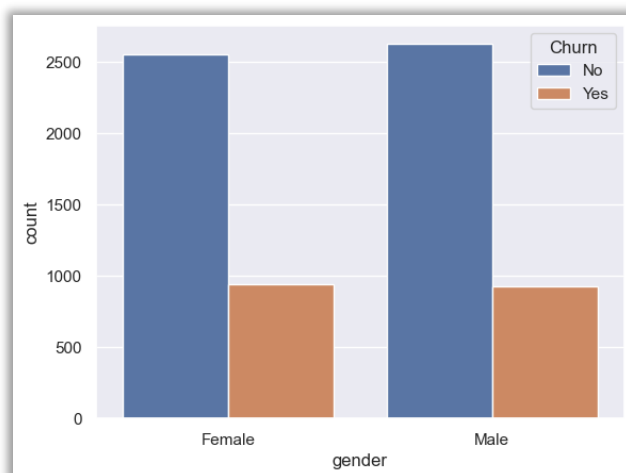
→ `sns.countplot(x='SeniorCitizen', hue='Churn', data=customer1)`



7. Gender vs Churn

- Churn distribution between male and female customers is nearly equal.
- Gender does not appear to be a strong predictor of churn.

→ `sns.countplot(x='gender', hue='Churn', data=customer1)`



8. Dependents and Partner Status

- Customers without dependents have higher churn rates compared to those with dependents.
- Customers without partners also show higher churn tendencies.
- Family association appears to reduce churn probability.

→ `pd.crosstab(index=customer1['Churn'], columns=customer1['Dependents'])`

Dependents	No	Yes
Churn		
No	3390	1784
Yes	1543	326

→ `pd.crosstab(index=customer1['Churn'], columns=customer1['Partner'])`

Partner	No	Yes
Churn		
No	2441	2733
Yes	1200	669

9. Payment Method vs Churn

- Customers using **Electronic Check** have the highest churn rate (~45%).
- Automatic payment methods (bank transfer and credit card) show significantly lower churn rates (~15–17%).
- Payment convenience and reliability may influence retention.

→ `pd.crosstab(customer1['PaymentMethod'], customer1['Churn'], normalize='index')`

Churn	No	Yes
PaymentMethod		
Bank transfer (automatic)	0.832902	0.167098
Credit card (automatic)	0.847569	0.152431
Electronic check	0.547146	0.452854
Mailed check	0.808933	0.191067

10. Technical Support vs Churn

- Customers without tech support show higher churn (~41%).
- Customers with tech support have significantly lower churn (~15%).
- Access to technical support improves retention.



```
pd.crosstab(customer1['TechSupport'], customer1['Churn'], normalize='index')
```

Churn	No	Yes
TechSupport		
No	0.583645	0.416355
No internet service	0.925950	0.074050
Yes	0.848337	0.151663

11. Internet Service Type

- Fiber optic customers show higher churn (~42%) compared to DSL (~19%).
- Customers with no internet service have very low churn (~7%).
- Service quality or pricing differences may impact churn behavior.

→ `pd.crosstab(customer1['InternetService'],customer1['Churn'],normalize='index')`

Churn	No	Yes
InternetService		
DSL	0.810409	0.189591
Fiber optic	0.581072	0.418928
No	0.925950	0.074050

12. Additional Services (Security, Backup, Protection, Streaming)

- Customers without online security, backup, or device protection have higher churn rates.
- Customers subscribed to additional services generally show better retention.
- “No internet service” category consistently shows the lowest churn.

Key EDA Insight

Churn is strongly associated with:

- Short tenure
- Month-to-month contracts
- Higher monthly charges
- Electronic check payment method
- Lack of tech support and security services

Logistic Regression Model

After completing Exploratory Data Analysis, Logistic Regression was implemented to classify customers into churn (1) and non-churn (0) categories.

1. Data Preparation

- A deep copy of the original dataset was created to preserve the raw data.
- The target variable **Churn** was converted into numerical format:
 - No → 0
 - Yes → 1
- The `customerID` column was removed since it does not contribute to prediction.
- All categorical variables were converted into numerical format using one-hot encoding (Dummy variables).
- Dummy variable trap was avoided by dropping the first category in each encoded feature.
- Independent variables (X) and dependent variable (y) were separated.

2. Train-Test Split

- The dataset was split into:
 - 70% Training Data
 - 30% Testing Data
- A fixed random state was used to ensure reproducibility.

3. Model Building

- Logistic Regression was initialized with:
 - `max_iter = 2000` to ensure convergence.
 - `class_weight = 'balanced'` to handle class imbalance in churn data.
- The model was trained using the training dataset.

4. Model Performance

Accuracy

- Model Accuracy: **74.63%**
- The model correctly classified approximately 75% of customers.

Misclassified Samples

- Total misclassified samples: **536**

Confusion Matrix

	Predicted No	Predicted Yes
Actual No	1138	422
Actual Yes	114	439

Interpretation:

- 1138 customers were correctly predicted as non-churners.
- 439 customers were correctly predicted as churners.
- 422 non-churners were incorrectly predicted as churners.
- 114 churners were missed by the model.

Classification Report Insights

- Precision (Churn): **0.51**
- Recall (Churn): **0.79**
- F1-Score (Churn): **0.62**

The recall value of 79% indicates that the model successfully identifies most customers who are likely to churn. However, the precision value suggests that some non-churn customers are incorrectly classified as churn.

5. Conclusion

- Logistic Regression performs reasonably well with ~75% accuracy.
- The model is particularly strong at identifying actual churners (high recall).
- Although some false positives exist, the model is effective for churn prediction where identifying potential churners is more critical than avoiding false alarms.

Customer Retention Strategies

Based on the insights obtained from Exploratory Data Analysis and Logistic Regression modeling, several key risk factors influencing customer churn were identified. The following business strategies are recommended to reduce churn and improve customer retention.

1. Contract-Based Retention Strategy

Observation: Customers with month-to-month contracts showed higher churn rates compared to long-term contract holders.

Strategy:

- Encourage customers to switch to 1-year or 2-year contracts through:
 - Discounted pricing
 - Loyalty benefits
 - Free service upgrades
- Offer contract renewal incentives before contract expiry.

2. Target Short-Tenure Customers

Observation: Customers with low tenure are more likely to churn.

Strategy:

- Introduce a structured onboarding program for new customers.
- Provide welcome benefits within the first 3 months.
- Offer proactive customer support during early service stages.
- Monitor new customers more closely using churn prediction alerts.

3. Manage High Monthly Charges

Observation: Customers with higher monthly charges have a higher probability of churn.

Strategy:

- Provide flexible pricing plans.
- Offer customized bundles based on usage.
- Provide periodic discount offers to high-paying customers.
- Introduce loyalty-based pricing reductions over time.

4. Improve Service Support

Observation: Customers without tech support or additional services are more likely to churn.

Strategy:

- Offer bundled packages including tech support.

- Provide quick-response customer service.
- Implement 24/7 support options.
- Use customer feedback systems to improve service quality.

5. Payment Method Optimization

Observation: Certain payment methods (e.g., electronic check users) show higher churn rates.

Strategy:

- Encourage automatic payment methods like credit card or bank transfer.
- Offer cashback or rewards for auto-pay enrollment.
- Simplify billing transparency to reduce confusion.

6. Early Churn Prediction Alerts

Using the Logistic Regression model:

- Customers predicted with high churn probability can be flagged.
- Retention teams can contact these customers proactively.
- Offer personalized discounts or retention packages.
- Monitor high-risk customers continuously.

7. Data-Driven Retention Programs

The predictive model allows businesses to:

- Segment customers based on churn risk.
- Allocate marketing budget efficiently.
- Reduce revenue loss by focusing on high-risk groups.

Overall Business Impact

Implementing these strategies can:

- Reduce customer acquisition costs.
- Increase customer lifetime value (CLV).
- Improve brand loyalty.
- Enhance long-term revenue stability.

By combining data analysis with targeted retention strategies, businesses can transform churn prediction into actionable business decisions.

Conclusion

This project focused on analyzing customer churn behavior and identifying the key factors influencing customer attrition. Through detailed Exploratory Data Analysis (EDA), meaningful patterns and relationships were discovered between customer characteristics and churn status.

The analysis revealed that factors such as contract type, payment method, tenure, monthly charges, and additional service features have a stronger impact on churn compared to demographic attributes like gender or senior citizen status. Customers with shorter tenure, month-to-month contracts, and higher monthly charges were more likely to churn.

A Logistic Regression model was implemented to classify customers into churn and non-churn categories. The model achieved an overall accuracy of approximately **75%**, demonstrating reasonable predictive performance. Importantly, the model showed strong recall for churn customers (79%), meaning it successfully identified most customers who are likely to leave.

Although the model produces some false positives, in a real business scenario this is acceptable because identifying potential churners early allows companies to implement retention strategies such as personalized offers, improved customer support, or targeted engagement campaigns.

Overall, this project demonstrates how data analysis and predictive modeling can help businesses:

- Understand customer behavior
- Identify risk factors associated with churn
- Make data-driven decisions
- Improve customer retention strategies

Future improvements could include trying advanced models, feature engineering, and hyperparameter tuning to further enhance predictive performance.

THANK YOU